# Lab 1: System Calls in C

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## 1 Question 1: File Handling Using System Calls

**Problem Statement:** Write a C program using system calls to:

- Create/Open a file `"data"` (read & write, create if not exists, truncate if exists).
- Write user input to the file and close it.
- Reopen, read, and display the file content.
- Handle errors using `perror()`.

**Example Output:**

```
Enter a string: Hello, System Calls!
Content read from file:
Hello, System Calls!
```

## 2 Question 2: File Positioning with `lseek()`

**Problem Statement:** Write a C program using system calls to:

- Open a file named `"testfile.txt"` in read-only mode using `open()`.
- Read the first 19 bytes from the file and display them.
- Move the file pointer 10 bytes from the beginning using `lseek()`.
- Read the next 19 bytes from the new position and display them.

- Implement error handling for all system calls (`open`, `read`, `lseek`, `close`).

**System Calls to Use:** `open()`, `read()`, `lseek()`, `close()`
**Example Output (Assuming File Content: "ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890")**

```
First read: ABCDEFGHIJKLMNOPQRS
Second read after lseek: KLMNOPQRSTUVWXYZ1
```

# 3 Question 3: Directory Operations Using System Calls

**Problem Statement:** Write a C program using system calls to perform directory operations based on user choice. The program should:

- Display a menu with the following options:
  1. Create Directory
  2. Remove Directory
  3. Read Directory Contents
- Take user input for the selected operation.
- Perform the selected operation:
  - Create a directory using `mkdir()`.
  - Remove a directory using `rmdir()`.
  - Open and read a directory using `opendir()` and `readdir()` to list its contents.
- Implement error handling using `perror()` for system calls (`mkdir`, `rmdir`, `opendir`, `readdir`).
- Close the directory stream using `closedir()` after reading its contents.

**System Calls to Use:** `mkdir()`, `rmdir()`, `opendir()`, `readdir()`, `closedir()`
**Example Output:**

```
** MENU **
1. Create Directory
2. Remove Directory
3. Read Directory
Enter your choice: 1
Enter directory name: mydir
Directory 'mydir' created successfully.

** MENU **
1. Create Directory
2. Remove Directory
3. Read Directory
Enter your choice: 3
Enter directory name to open: mydir
Contents of directory 'mydir':
.
..
file1.txt
file2.txt
```